Insulin, Glucagon, and the Logic of Blood Sugar

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> Blood glucose is held within a narrow band by two hormones pulling in opposite directions. Insulin lowers it; glucagon raises it. This paper reads their interplay as a control system, and shows how diabetes is best understood as that control loop coming apart.

The body keeps blood glucose within a remarkably tight range — roughly four to seven millimoles per litre — despite meals that flood the blood with sugar and fasts that drain it. It manages this with a pair of opposing hormones secreted by the islets of Langerhans in the pancreas: insulin, from the beta cells, and glucagon, from the alpha cells.

Two hormones, opposite signs

After a meal, glucose rises. The beta cells sense it and release insulin. Insulin is the “store it” signal: it tells muscle and fat to pull glucose out of the blood, and tells the liver to lock glucose away as glycogen. Blood sugar falls back toward baseline.

During a fast, glucose falls. Now the alpha cells release glucagon — the “release it” signal. Glucagon instructs the liver to break glycogen back down into glucose and to manufacture new glucose from other fuels. Blood sugar rises back toward baseline.

> *Neither hormone sets the level alone. The stable number in the blood is the running balance of two opposing pushes — a set-point maintained, not a value dialled in.*

A feedback loop, drawn plainly

The system is a textbook negative-feedback loop. The regulated variable is blood glucose. The sensors are the pancreatic islet cells. The effectors are the liver, muscle, and fat. A deviation in either direction triggers the hormone that opposes it, and the opposition is what produces stability.

- **Glucose high** → insulin up, glucagon down → storage.
- **Glucose low** → glucagon up, insulin down → release.

Diabetes as a broken loop

Seen this way, the two main forms of diabetes are two distinct failures of the same loop. In type 1 diabetes, the beta cells are destroyed by the immune system: the “store it” signal is gone, so glucose rises unchecked and must be replaced from outside. In type 2 diabetes, the signal is present but the tissues have grown deaf to it — insulin resistance — so the pancreas shouts louder until, eventually, it tires.

The clinical picture in each case follows directly from the diagram: the loop cannot bring glucose down, and the consequences accumulate wherever chronically high glucose does its slow damage — vessels, nerves, kidneys, eyes. To understand the disease is, in large part, to understand the control system it disrupts.
